

FITS047

Assignment 2: Knowledge Representation and Reasoning

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Q1 a) Proposition

$$\phi_1 = A \rightarrow ((A \rightarrow B) \vee (B \rightarrow C))$$

$$\Rightarrow A \rightarrow ((\neg A \vee B) \vee (\neg B \vee C)) \quad [\text{using implication elimination}]$$

$$\Rightarrow \neg A \vee ((\neg A \vee B) \vee (\neg B \vee C))$$

$$\Rightarrow \neg A \vee (\neg A \vee B \vee \neg B \vee C)$$

$$\Rightarrow \neg A \vee \neg A \vee B \vee \neg B \vee C$$

$$\Rightarrow \neg A \vee B \vee C$$

As this sentence can not always be true, it is satisfiable but not valid.

$$\phi_2 = (A \wedge B) \leftrightarrow ((B \rightarrow C) \rightarrow ((C \wedge A) \wedge A))$$

→ Using biconditional elimination

$$(A \wedge B) \rightarrow ((B \rightarrow C) \rightarrow ((C \wedge A) \wedge A)) \wedge ((B \rightarrow C) \rightarrow ((C \wedge A) \wedge A)) \rightarrow (A \wedge B)$$

→ not remove implication

$$\neg(A \wedge B) \vee ((B \rightarrow C) \rightarrow ((C \wedge A) \wedge A)) \wedge \neg((B \rightarrow C) \rightarrow ((C \wedge A) \wedge A)) \vee (A \wedge B)$$

$$\neg(A \wedge B) \vee (\neg B \vee C) \vee ((C \wedge A) \wedge A) \wedge \neg((B \rightarrow C) \rightarrow ((C \wedge A) \wedge A)) \vee (A \wedge B)$$

Now using truth table

A	B	C	$A \wedge B$	$B \rightarrow C$	$C \wedge A$	$(C \wedge A) \vee A$	$(B \rightarrow C) \rightarrow ((C \wedge A) \vee A)$	Φ_2
F	F	F	F	F	F	F	T	T
F	F	T	F	T	F	F	F	T
F	T	F	F	F	F	F	T	T
F	T	T	F	T	F	F	F	T
T	F	F	F	T	F	T	T	T
T	F	T	F	T	T	T	T	T
T	T	F	T	F	F	T	T	T
T	T	T	T	T	T	T	T	T

As all combinations are True

Conclusion = It is a valid statement

$$\Phi_3 = (A \leftrightarrow B) \wedge (\neg(A \wedge \neg C) \vee (B \leftrightarrow D))$$

→ Using biconditional

$$(A \leftrightarrow B) = (\neg A \vee B) \wedge (\neg B \vee A)$$

$$(B \leftrightarrow D) = (\neg B \vee D) \wedge (\neg D \vee B)$$

→ Using conjunction De Morgan

$$\neg(A \wedge \neg C) \approx \neg A \vee C$$

$$\therefore (\neg A \vee B) \wedge (\neg B \vee A) \wedge (\neg A \vee C) \vee ((\neg B \vee D) \wedge (\neg D \vee B))$$

Using truth table

A	B	C	D	$A \oplus B$	$B \leftrightarrow D$	$A \neg C$	$\neg(A \neg C)$	Φ_3
F	F	F	F	T	T	F	T	T
F	F	F	T	T	F	F	T	T
F	F	T	F	T	T	F	T	T
F	F	T	T	T	F	F	T	T
F	T	F	F	F	F	F	T	F
F	T	F	T	F	T	F	T	F
F	T	T	F	F	F	F	T	F
F	T	T	T	F	T	F	T	F
T	F	F	F	F	T	T	F	F
T	F	F	T	F	F	T	F	F
T	F	T	F	F	T	F	T	F
T	F	T	T	F	F	F	T	F
T	T	F	F	T	F	T	F	F
T	T	F	T	T	T	F	T	T
T	T	T	F	T	F	T	T	T
T	T	T	T	T	F	T	T	T

$\therefore \Phi_3$ is satisfiable not Valid.

6) Checking Horn Clause, CNF form done earlier.

Φ_1 :- Not a Horn clause as it contains multiple positive literals

Φ_2 :- Not a Horn clause as it contains multiple positive literals

Φ_3 :- Not a Horn clause as it contains multiple positive literals

Q2 Forward/Backward Reasoning

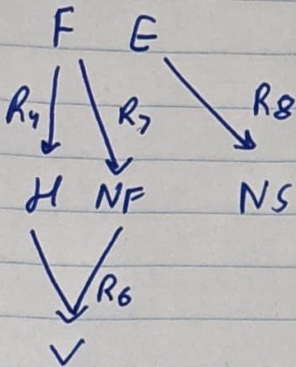
1. $(V \wedge P \wedge T) \rightarrow S$
2. $\neg H \vee \neg NS \vee T$
3. $\neg P \rightarrow (\neg NF \vee \neg NS)$
4. $F \rightarrow H$
5. $P \rightarrow W$
6. $(H \wedge NF) \rightarrow V$
7. $NF \vee \neg F \quad (F \rightarrow NF)$
8. $\neg NS \rightarrow \neg E \quad (E \rightarrow NS)$

Initial facts :-

$F \Rightarrow \text{true}$

$E \Rightarrow \text{true}$

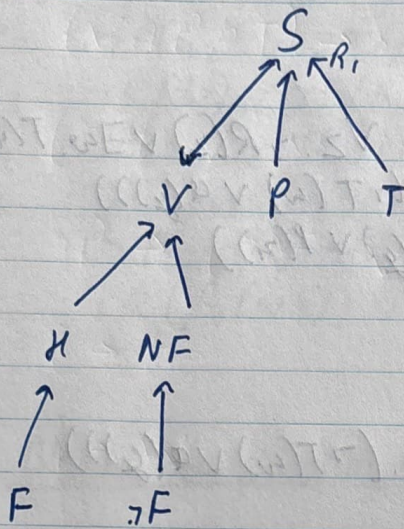
Forward Chaining



Queue	Count								Inferred
	R ₁	R ₂	R ₃	R ₄	R ₅	R ₆	R ₇	R ₈	
F, E	1	3	2	1	1	1	1	1	
E, NF, H	1	3	2	0	1	1	0	1	F
NF, H, NS	1	3	2	0	1	1	0	0	F, G
V, NS	1	3	2	0	1	0	0	0	F, G, NF, H

As the state of P, T is not getting proved by the forward chaining. There are not enough facts to prove the statement.

Backward Chaining



$\therefore$ Here Backward chaining can't prove S because it can't prove P and T

Q3. First order logic

1) Constructing Clauses

2. Eliminate Implications:

- Rule 1: $\neg(\exists z. R(z)) \rightarrow (\exists w. T(w)) \approx \forall z. \neg R(z) \vee \exists w. T(w)$
→ Rule 2: $\forall x \exists y. \forall z. (\neg(P(x) \vee R(z)) \vee \exists w. (\neg T(w) \vee Q(y)))$
→ Rule 3: $\forall w \exists y. \forall x (\neg P(x) \vee T(w)) \wedge \forall z. (\neg Q(y) \vee P(z))$

2. Move negation inwards:

- Rule 2: $\forall x \exists y. \forall z. ((\neg P(x) \wedge \neg R(z)) \vee \exists w. (\neg T(w) \vee Q(y)))$

3. Eliminate Existential Quantifiers:

- Rule 1: $\forall z. \neg R(z) \vee T(f_1(z))$
Rule 2: $\forall x \forall z. ((\neg P(x) \wedge \neg R(z)) \vee (\neg T(f_2(z, z)) \vee Q(f_3(z, z))))$
Rule 3: $\forall w \forall x. (\neg P(x) \wedge T(w)) \wedge \forall z. (\neg Q(f_4(w)) \vee P(z))$

Skolem functions: f_1, f_2, f_3, f_4

4. Put into CNF form:

- Rule 1: $\forall z. \neg R(z) \vee T(f_1(z))$
Rule 2: $\forall x \forall z. (\neg P(x) \vee \neg T(f_2(z, z)) \vee Q(f_3(z, z))) \wedge (\neg R(z) \vee \neg T(f_2(z, z)) \vee Q(f_3(z, z)))$
Rule 3: $\forall w \forall x. (\neg P(x) \vee T(w)) \wedge \forall z. (\neg Q(f_4(w)) \vee P(z))$

5. Eliminate universal quantifiers:

- Rule 1: $\neg R(z) \vee T(f_1(z))$
Rule 2: $(\neg P(x) \vee \neg T(f_2(z, z)) \vee Q(f_3(z, z))) \wedge (\neg R(z) \vee \neg T(f_2(z, z)) \vee Q(f_3(z, z)))$
Rule 3: $(\neg P(x) \vee T(w)) \wedge (\neg Q(f_4(w)) \vee P(z))$

6. Renaming variables

- $\rightarrow C_1: \neg R(z_1) \vee T(f_1(z_1))$
- $C_2: \neg P(x_2) \vee \neg T(f_2(x_2, z_2)) \vee Q(f_3(x_2, z_2))$
- $C_3: \neg R(z_3) \vee \neg T(f_2(x_3, z_3)) \vee Q(f_3(x_3, z_3))$
- $C_4: \neg P(x_4) \vee T(w_1)$
- $C_5: \neg Q(f_4(w_2)) \vee P(x_5)$
- $C_6: \neg P(x_6) \vee T(w_3)$
- $C_7: \neg Q(f_4(w_4)) \vee P(x_7)$

7. Negate the goal and add to the set of clauses:

- $\rightarrow$ Negated Goal: $\neg(\forall x. P(x))$ which becomes $\exists x. \neg P(x)$
- $\rightarrow$ Skolemize: $\neg P(c)$
- $\rightarrow C_8: \neg P(c)$

b) Now Applying the proof method

1. Resolve $C_8: \neg P(c)$ with $C_4: \neg P(x_4) \vee T(w_1)$
 - Unifier: $\{x_4/c\}$
 - Result: $C_9: T(w_1)$
2. Resolve $C_9: T(w_1)$ with $C_1: \neg R(z_1) \vee T(f_1(z_1))$
 - Unifier: $\{w_1/f_1(z_1)\}$
 - Result: $C_{10}: \neg R(z_1) \vee T(f_1(z_1))$
3. Resolve $C_8: \neg P(c)$ with $C_5: \neg Q(f_4(w_2)) \vee P(x_5)$
 - Unifier: $\{x_5/c\}$
 - Result: $C_{11}: \neg Q(f_4(w_2))$
4. Resolve $C_9: T(w_1)$ with $C_2: \neg P(x_2) \vee \neg T(f_2(x_2, z_2)) \vee Q(f_3(x_2, z_2))$
 - Unifier: $\{w_1/f_2(x_2, z_2)\}$
 - Result: $C_{12}: \neg P(x_2) \vee Q(f_3(x_2, z_2))$

5. Resolve $C8: \neg P(c)$ with $C12: \neg P(x_2) \vee q(f_3(x_2, z_2))$

Unifier: $\{x_2/c\}$

Result: $C13: q(f_3(c, z_2))$

6. Resolve $C11: \neg q(f_4(w_2))$ with $C13: q(f_3(c, z_2))$

Unifier: $\{f_4(w_2)/f_3(c, z_2)\}$

Result: None

c) Most General Unifiers (MGUs)

* 1: $\{x_1/c\}$

2: $\{w_1/f_1(z_1)\}$

3: $\{x_5/c\}$

4: $\{w_1/f_2(x_2, z_2)\}$

5: $\{x_2/c\}$

6: $\{f_4(w_2)/f_3(c, z_2)\}$

As the negated goal is unsatisfiable

$\therefore$ the original goal is provable.

* Generative AI has been used in a guided ~~not~~ manner to help with the understanding of the Assignment structure